MILITARY SPECIFICATION

CONVERTERS, AIRCRAFT, DC TO DC, BOOST GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers general requirements for converters for use in converting electrical power from 9 to 28 volts dc to regulated dc power at 28 volts dc.

1.2 Classification. Converters shall be classified by Class as specified in 1.2.1.

1.2.1 Class. Converters shall be of the following classes, as specified by the temperature-altitude characteristics:

Class A. Class A converters shall be designed to meet the temperature-altitude requirements of MIL-E-81910(AS) for self-cooled class A equipment.

Class B. Class B converters shall be designed to meet the temperature altitude requirements of MIL-E-81910(AS) for self-cooled class B equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal	
QQ-P-416	Plating, Cadmium (Electrodeposited)
Military	
MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy; Processes for Corrosion, Protection of
MIL-W-5088	Wiring, Aircraft, Selection and Installation Of Electronic Equipment, Airborne, General Specification For

FSC 6130

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SPECIFICATIONS (Cont'd)

Military

MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-I-45208	Inspection System Requirements
MIL-E-81910(AS)	Electric Power and Control Equipment, Aircraft, Testing, General Specification For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order Of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Characteristics and Utilization of Aircraft Electric Power
MIL-STD-749	Preparation and Submission of Data for Approval of Non-Standard Electronic Parts
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Report, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of

Federal

FED-STD-595	Color

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification. The converter furnished under this specification shall be a product that has been tested and passed the qualification tests herein and has been listed on or approved for listing on the applicable qualified products list (see 6.2).

3.2 Detail specifications. The individual converter requirements shall be as specified herein and in accordance with the applicable detail specification sheets, or MS sheets. In the event of any conflict between requirements of this specification and the detail specification sheet, or MS sheet, the latter shall govern.

3.3 Materials.

3.3.1 Metals. Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

3.3.1.1 Dissimilar metals. Unless otherwise suitably protected, dissimilar metals, such as brass, copper, or steel, shall not be used in intimate contact with magnesium, aluminum or their alloys. When protection is used, it shall be of such type that a low-impedance path is offered to radio-frequency currents. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Non-metallic materials. Non-metallic materials used shall be flame resistant, shall not support combustion, be non-toxic and fungus inert.

3.3.3 Finishes and protective coatings.

3.3.3.1 Aluminum alloy parts. Aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the actual contact area of all surfaces that act as a path for electrical current and from the areas under screws, nuts, etc., used for assembly or mounting purposes.

3.3.3.2 Magnesium alloy parts. Magnesium alloy parts shall not be used without approval of the qualifyingaactivity. When approved for use, magnesium alloy shall be surface treated in accordance with MIL-M-3171 to provide protection against corrosion.

3.3.3.3 Steel parts. Steel parts shall be cadmium plated in accordance with Type II or III of QQ-P-416.

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3.3.3.4 Color. The converter shall be finished in 17875 white in accordance with Federal Standard 595.

3.4 Selection of parts and materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the design and construction of converters complying with the requirements of this specification shall be selected in accordance with Standard MIL-STD-143.

3.4.1 Standard parts. Military standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Any standard parts of the same number, shall be readily interchangeable as applied within the converter. In the event there is no suitable standard part available as of the date of invitation for bids, a non-standard part may be used. Integrated circuits, if utilized shall be rated for operation to 125°C minimum. Transistors shall be metal cased type and rated for operation to 150°C minimum. Capacitors shall be rated for operation at 105°C minimum.

3.4.2 Non-standard Parts. The use of non-standard parts shall be approved by the activity responsible for qualification. Request for approval for the use of non-standard parts shall be submitted in accordance with MIL-STD-749.

3.4.3 Commercial (hardware) parts. Commercial utility hardware items such as screws, bolts, nuts and cotter pins may be used, provided they possess suitable properties and are replaceable by MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and on the contractor's drawings.

3.5 Design and construction. The converter shall be designed as a single package and shall be a completely static solid state unit having no moving parts. No vacuum or gas filled tubes shall be used. Protective devices (fuses, circuit breakers and thermal actuated) shall not be used.

3.5.1 Mechanical.

3.5.1.1 Maintainability. The converter shall be designed to facilitate assembly, disassembly, location of trouble sources,

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and maintenance without the aid of special tools. All functional parts shall be readily identifiable, accessible and removable for replacement. When printed circuit boards are utilized, only clear transparent conformal coating shall be used in order to permit identification, location, removal and replacement of functional parts. Functional parts are defined as individual component parts such as resistors, capacitors, semiconductor devices, inductive devices or replaceable functional assemblies such as active or passive filters, voltage dividers, dual transistors or diodes.

3.5.1.2 Interchangeability. Provisions for interchangeability are required and shall be in accordance with Requirement 7 of MIL-STD-454.

3.5.1.3 Screw threads. Screw threads shall conform to the requirements of MIL-S-7742.

3.5.1.4 Safety wiring and staking. Accidental loosening of screws and screw parts, and other connections, shall be prevented by safety wiring (0.032-inch minimum OD) where practicable, staking or other approved method. Washers and cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.5.1.5 Cooling. The converter shall be self-cooled by natural convection and radiation. No conduction cooling shall be allowed. Safe operating equilibrium temperature shall be maintained for the converter over the specified range of operating conditions, and the rated temperature of individual converter components shall not be exceeded.

3.5.1.6 Ventilation openings. All ventilation or other openings shall be designed to prevent passage of foreign objects and shall be of such size as to prevent passage of 5/16-inch diameter rod.

3.5.1.7 Operating position. The converter shall be capable of satisfactory operation in any position.

3.5.1.8 Size and weight. The physical dimensions, weight and envelope configuration shall be as specified (see 3.2). The weights and dimensions shown on the detail specifications shall include the weights and dimensions of all auxiliary apparatus necessary to make the converter conform to this specification and the detail specifications.

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3.5.2 Electrical.

3.5.2.1 Input. The converter shall be designed to meet the requirements of this specification when supplied with power from an aircraft electrical system, conforming to MIL-STD-704, for Category B equipments.

3.5.2.2 Influence on electric system. The converter shall not influence the DC power supply system in excess of the limits specified in the paragraph entitled "Influence on Electric System" in MIL-STD-704. The DC power supply system shall be a standard aircraft system with a rated capacity of at least five times the converter rating. Peak to mean input ripple currents shall not exceed the mean input current or the values specified below, whichever is lower, when the inverter is operated from a zero impedance source.

CONVERTER RATING-AMPERES	PEAK TO MEAN INPUT RIPPLE CURRENT MAX-AMP
Under 1	0.5
1 - 3	1
4 - 7	2
8 - 20	5
21 - 30	10

3.5.2.3 Adjustments. The converter shall have no external voltage adjustments nor shall any adjustment be necessary at time of installation. All adjustments shall be made at the time of manufacture or overhaul.

3.5.2.4 Electric connections.

3.5.2.4.1 External electric connections. The equipment shall be so designed that external connections and wiring can be made in accordance with MIL-W-5088. Equipment structure shall not be used as a current path except for radio noise shielding. Connectors shall be polarized or sized so as to prevent their interchange during equipment installation.

3.6 Performance. No warm-up time shall be allowed to meet the performance requirements.

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3.6.1 Operating conditions. The converter shall deliver power in accordance with the requirements specified herein and on the detail specification (see 3.2).

3.6.1.1 Output voltage regulation. Unless otherwise specified (see 3.2), the converter shall maintain the following steady state output voltages under all load, temperature and altitude conditions specified (see 4.7.2):

INPUT VOLTAGE	OUTPUT VOLTAGE
9 - 28.5 Volts DC	26 - 28.5 Volts DC
28.5 - 32 Volts DC	26 Volts DC minimum, output not to exceed input

3.6.1.2 Input Voltage vs Time: The converter shall be capable of operating sequentially for the following time periods at the indicated input voltage with output loads from zero to full load.

9 - 10 Volts DC for 1 Minute
10 - 11 Volts DC for 2 Minutes
11 - 12 Volts DC for 5 Minutes
12 Volts DC and Higher Continuously

3.6.1.3 Input power susceptibility. The converter shall not be damaged by the input surge and spike voltages defined in 4.7.11. There shall be no evidence of flashover or insulation breakdown as a result of the tests specified in 4.7.11, and the output transients generated during the test shall not exceed the limits defined in MIL-STD-704, Figure 9, Curve 11. During the test specified in 4.7.11.3, the output voltage and ripple of the converter shall be within the limits specified in 3.6.1.1, and 3.6.4.

3.6.2 Overload. Over the input voltage range of 20 to 32 VDC, the converter shall be capable of delivering the current indicated below without damage for the specified time period. The converter shall maintain the minimum specified output voltage indicated while delivering specified overload current.

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Output Current (percent of rated load current)	Minimum Output Voltage (Volts)	Time 20-27 VDC Input	Time 27-32 VDC Input
1000%	18	Not Required	5 Sec.
625%	20	Not Required	30 Sec.
375%	24	15 Sec.	1 Min.
250%	24	1 Min.	3 Min.
175%	25	5 Min.	15 Min.
150%	25	10 Min.	30 Min.
125%	26	20 Min.	60 Min.

3.6.3 Efficiency. The efficiency of the converter shall be within the following limits at full rated load.

INPUT VOLTAGE	EFFICIENCY (% min)
9 - 28 VDC	60
28 - 32 VDC	85

3.6.4 Output Ripple. Output ripple voltage shall meet the requirements of MIL-STD-704 at minimum, maximum and room ambient temperatures at sea level altitude.

3.6.5 Source Impedance: The converter shall meet all requirements of this specification with input power supplied from source impedances of zero to the value specified in the detail specification.

3.6.6 Recovery time. During application and removal of normal rated load, output voltage shall remain between 18 and 40 volts and shall recover to and remain in the range of 24 and 32 volts within 50 milliseconds.

3.6.7 Starting Voltage: Output voltage during starting shall be within the range of 24 and 32 volts within 0.2 seconds after application of input voltage. Output voltage shall not exceed 35 volts during starting.

3.6.8 Starting Current: Peak input starting current shall not exceed 75 amps and shall recover to within 10 percent of steady state within 25 milliseconds.

3.6.9 Electro-magnetic compatibility. When tested as specified in 4.7.7, the narrowband and broadband conducted and radiated electro-magnetic interference shall be within the limits specified in MIL-STD-461A, notice 1.

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3.7 Environmental service.

3.7.1 Temperature-Altitude. The converter shall be capable of operation without degradation of performance throughout the temperature-altitude range specified by MIL-E-81910(AS), for the class indicated in the detail specification.

3.7.2 Humidity. As a result of the humidity test (see 4.7.13) there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.6.1.1.

3.7.3 Dust. The converter shall be capable of operating within specified limits during the dust test (see 4.7.19). At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.6.1.1.

3.7.4 Salt fog. As a result of the salt fog test (4.7.18) there shall be no evidence of corrosion, peeling, pitting, blistering or cracking of the finish. At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.6.1.1.

3.7.5 Fungus. As a result of the fungus test (see 4.7.17), the converter shall show no evidence of fungus growth on any surface. At the conclusion of the test, the voltage regulation shall be within the limits specified in 3.6.1.1.

3.7.6 Vibration. The input voltage during the test shall be 20 volts and the converter shall be operated at full load. The output voltage measured during the test shall remain within the limits specified in 3.6.1.1. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test the voltage regulation shall be within the limits specified in 3.6.1.1.

3.7.7 Shock. The converter shall be operated at 20 volts input and full load during the basic design portion of the test. The output voltage shall remain within the limits specified in 3.6.1.1. during and following the basic design portion of the test.

3.7.8 Acceleration. The output voltage of the converter shall remain within the limits of 3.6.1.1 while operating at

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20 volts input and full load during the operating portion of acceleration test (see 4.7.12). There shall be no evidence of physical damage and the voltage regulation after test shall be within the limits specified in 3.6.1.1.

3.7.9 Explosive atmosphere. (as required - see 3.2). The converter shall not cause ignition when tested as specified in 4.7.15.

3.8 Reliability. The converter shall have a minimum acceptable MTBF (mean-time-between-failure) of 2000 hours, demonstrated to a 90 percent confidence level by the test specified in 4.8.

3.9 Markings.

3.9.1 Terminals. Markings of external electrical connections shall be as specified on the applicable detail specification sheet (see 3.2.)

3.9.2 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.9.3 Nameplate. A nameplate conforming with the following markings, in addition to the requirements of MIL-STD-130 shall be securely attached to the converter. The nameplate shall contain the following information:

Converter, DC to DC, Boost
Input: 9-32 Volts DC 18 Amperes DC
Output: 28 Volts DC 4 Amperes DC
Military Part No.
Federal Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trade-mark
Inspector's stamp
US

3.10 Workmanship. The converter shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

3.11 Installation instructions. The contractor shall pack with each converter one printed copy of simple instructions, with

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illustrations and required diagrams covering its installation. These instructions shall be protected to withstand and remain completely legible under the humidity and fungus requirements specified herein for self cooled components. Prior to printing, two copies shall be furnished the qualifying activity for approval of content, presentation, and means of attachment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system, which the supplier is required to maintain as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208.

4.2 Classification of inspection. The inspection and testing of the equipment shall be classified as follows:

a. Qualification inspections (see 4.3).

b. Quality conformance inspections (see 4.4).

c. Qualification verification inspection (see 4.5).

4.3 Qualification inspections.

4.3.1 Qualification tests. Qualification tests shall be conducted, at a facility satisfactory to the activity responsible for qualification (see 6.2). Qualification tests shall consist of the examination, inspections and tests specified in Table 1 - essentially in the order shown.

4.3.2 Failures. One or more failures may be cause for refusal to grant qualification approval. In the event of a failure during qualification, the manufacturer shall develop the necessary corrective action and retest procedures and plans. These shall be coordinated with the activity responsible for qualification. Corrective action, retest or continuation of tests shall not be initiated without the approval of the activity responsible for qualification.

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TABLE 1 - QUALIFICATION INSPECTION

TEST	CONVERTER SAMPLE 1	2	3	4	REFERENCE PARAGRAPH REQUIREMENT	TEST METHOD
Visual & mechanical inspection	X	X	X	X	3.3,3.4,3.5.1, 3.5.2.3, 3.5.2. 4.1, 3.9,3.10, 3.11	4.7.1
Voltage regulation					3.6.1.1, 3.6.1.2	4.7.2
Room ambient temperature	X	X	X	X		4.7.2.1
Maximum temperature-altitude	X	X	X	X		4.7.2.2
Minimum temperature-altitude	X	X	X	X		4.7.2.3
Operating position	X	X				4.7.2.4
Worst condition	X	X				4.7.2.5
Efficiency	X	X	X	X	3.6.3	4.7.3
Recovery Time			X	X	3.6.6	4.7.4
Starting		X		X	3.6.7, 3.6.8	4.7.5
Overload	X	X	X	X	3.6.2	4.7.6
Electro-magnetic compatibility	X		X		3.6.9	4.7.7
Output Ripple Voltage	X	X	X	X	3.6.4	4.7.8
Influence on electrical system	X	X	X	X	3.5.2.2	4.7.9
Temperature-altitude cycling	X	X			3.7.1	4.7.10
Input power susceptibility	X	X			3.6.1.3, 3.6.5	4.7.11
Acceleration	X				3.7.8	4.7.12
Humidity	X				3.7.2	4.7.13
Shock	X				3.7.7	4.7.14
Explosive atmosphere (when specified)			X		3.7.9	4.7.15
Vibration		X			3.7.6	4.7.16
Fungus		X			3.7.5	4.7.17
Salt fog		X			3.7.4	4.7.18
Dust				X	3.7.3	4.7.19
Reliability			X	X	3.8	4.8

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4.3.3 Qualification samples. The qualification test sample shall consist of at least four units of each part number for which qualification is sought. The sample units shall be produced, to the maximum extent practicable, with equipments and procedures normally used in production, and shall be identified as specified (see 3.9) and any information required by the letter of authorization (see 6.2).

4.3.4 Data submission.

4.3.4.1 Drawings and instructions. Together with the test samples, the manufacturer shall submit one reproducible copy of the outline and assembly drawing, schematic, parts list and two printed copies of simple instructions with illustrations and diagrams, if necessary, covering the installation of the converter.

4.3.4.2 Materials certification. Together with the test samples, the manufacturer shall submit a materials inspection report consisting of certification, supported by verifying data, that the materials and finishes specified in 3.3, used in fabricating the converters are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.3.5 Disposition of qualification samples and test report. Upon completion of the qualification tests, the test samples shall be identified as required and forwarded to the activity responsible for qualification (see 6.2). The test samples shall be accompanied by three copies of the qualification test report, prepared in accordance with MIL-STD-831.

4.3.6 Retention of qualification. To retain qualification, the manufacturer shall comply with the requirement of paragraph 4.5, Qualification verification. In the event that no production occurred during a 12-month period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product for testing in accordance with the qualifiation inspection requirements.

4.4 Quality conformance inspections. Quality conformance inspections shall consist of inspection of product for the delivery (see 4.4.1).

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4.4.1 Inspection of product for delivery. Inspection of product for delivery against a contract or purchase order shall consist of individual inspections (see 4.4.1.1).

4.4.1.1 Individual inspection. Each converter offered for delivery against a contract or purchase order shall be subjected to the tests listed in Table 2, in the sequence shown.

4.4.1.2 Failure. Any converter failing to comply with specified requirements during the individual inspections shall not be delivered on the contract or purchase order. The manufacturer may take corrective action on the failed converters as appropriate and resubmit for inspection. Reworked units shall be suitably marked to identify them as repaired items. In the event of failure after reinspection, further rework and resubmission shall not be allowed.

TABLE 2 - INDIVIDUAL INSPECTIONS

TEST	REFERENCE PARAGRAPH	
	REQUIREMENT	TEST METHOD
Visual and mechanical inspection	3.3, 3.4, 3.5.1, 3.5.2.3, 3.5.2.4.1, 3.9, 3.10, 3.11	4.7.1
Burn-in	--	4.8.3, 4.8.4
Voltage regulation Room ambient temperature	3.6.1.1, 3.6.1.2	4.7.2.1
Efficiency	3.6.3	4.7.3
Output ripple voltage	3.6.4	4.7.8

4.4.1.3 Non-compliance. In the event that more than 5 percent of the converters offered for delivery fail to pass the individual inspections, the entire lot will be considered to have failed. Acceptance of product shall be discontinued until corrective action acceptable to the activity responsible for qualification, has been taken.

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4.5 Qualification verification. Each converter of the same part number delivered as a qualified item under this specification shall be numbered sequentially in essentially the order that it is submitted for Government acceptance regardless of the contract under which delivered or by whom purchased. To meet the delivery requirement for the first 50 converters, 51 converters must successfully complete the required acceptance tests; 50 of which are to be shipped as directed by the applicable contracts, and one converter selected at random by the Government Inspector is to be forwarded to the Naval Air Test Center (SY61), Patuxent River, Maryland (Attention: NATC-8P7-101). To meet the delivery for each succeeding 150 converters, 151 converters must successfully complete the required acceptance test, 150 of which are to be shipped as directed by the applicable contract(s) and one converter selected at random and forwarded as above. The span of serial numbers of the production units from which each verification sample was selected shall be recorded and this information included with the sample. The sample shall be shipped no later than the final unit in the production group from which selected. The units forwarded to Naval Air Test Center will be inspected and tested for conformance to the applicable detail specification. Tests will be conducted as shown by Table 1 for any one of the qualification samples at the discretion of the qualifying activity. The sample unit will be returned to the contractor with a report of the Government test results at the conclusion of the design verification tests. Delivery of production converters is not to be delayed pending result of any tests conducted on qualification verification samples. Once qualified, production converters successfully passing the acceptance tests are to be accepted until qualification is officially rescinded. The requirements of Provisions Governing Qualification Testing, SD-6, Part II, shall pertain. Shipping authorization and instructions are to be given by the cognizant Government Inspector when the sample unit is selected. The price of testing to be borne by the contractor shall be fixed by the price list issued by the Naval Air Systems Command (AIR-5364), of the issue in effect on the date of the invitation for the bid, which led to the production associated with the test sample. If the test sample is selected from a group of production units involving more than one contract, the price for testing will be determined by the date of the bid associated with the oldest of these contracts. A certified check (made payable to Commander, Naval Air Test Center) for the full amount of the testing is to be submitted with each unit.

4.6 Test conditions. Unless otherwise specified, tests shall be made under the following conditions.

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4.6.1 Ambient temperature. Tests shall be performed at an ambient temperature of 25° ± 15°C.

4.6.2 Ambient altitude. Tests shall be run at approximately sea level altitude.

4.6.3 Tolerance and accuracy. The tolerance of test conditions, other than ambient, and the accuracy of test apparatus shall be as specified in MIL-E-81910(AS).

4.6.4 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperature, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above, the converter, and averaging the results. Air temperature measurements shall be taken 1 inch ± 1/4 inch from the surface of the converter.

4.6.5 Air flow. For all high temperature-altitude tests below 20,000 feet, the air flow over the converter within the test chamber shall be adjusted to approximate natural convection conditons and shall not exceed 100 feet/minute while maintaining the required temperature within specified tolerance. For tests above 20,000 feet altitude, the air flow may exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required temperature within specified tolerance.

4.6.6 Stabilization. Stabilization of test conditions is defined to have occurred when the change, over a ten minute period is measured to be less than:

a. Voltage - .01 volts

b. Converter critical component temperature, 1°C

4.6.7 Voltage measurement. The input and output voltage shall be measured at the terminals of the unit.

4.6.8 Position. The units shall be mounted base down.

4.6.9 Loading.

4.6.9.1 Definition of full load. Full load shall be the maximum continuous load specified at the applicable input voltage and temperature-altitude conditions (see 3.2).

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4.7 Test and inspection methods.

4.7.1 Visual and mechanical inspection. The converter shall be inspected to determine compliance with the requirements of this specification and the applicable detail specification sheet (see 3.2) with respect to materials, design, and construction parts, workmanship, dimensions, weight and markings, for which there are no applicable tests.

4.7.2 Voltage and frequency regulation. The output voltage shall be measured under the conditions specified in 4.7.2.1 through 4.7.2.5. For each load, input voltage and temperature-altitude condition, output voltage shall remain within the limits specified (see 3.6.1.1, and 3.2). During the course of the tests no components of the converters shall exceed their maximum rated temperatures. Tests may be conducted at any intermediate input voltage, load, or temperature-altitude condition at the option of the activity responsible for qualification.

4.7.2.1 Voltage regulation at room ambient temperature. The following tests shall be conducted immediately upon energizing the converter and then repeated after operating the converter at 12 volts input, full load, until output voltage is compleletely stabilized. The input voltage shall be varied in one volt increments from 9 to 12 volts and from 12 to 32 volts in two volt increments. At each input voltage, the load shall be varied from no load to full rated load in steps of one-fourth of rating. At each input voltage and load condition, the converter shall demonstrate its ability to start. For input voltages between 9 and 12 volts the full load conditions shall be allowed to remain for the time specified in 3.6.1.2.

4.7.2.2 Voltage regulation at maximum temperature-altitude conditon. The converter shall be tested in accordance with MIL-E-81910(AS) for the particular class of equipment (see 1.2.1 and 3.2). Prior to being energized, the converter shall be soaked, non-operating, at the initial maximum temperature, sea level condition for at least one-half hour. The test of 4.7.2.1 shall then be conducted except that critical component temperature shall also be considered in determining stabilization. For subsequent temperature-altitude conditions, voltage regulation tests need only be conducted following stabilization.

4.7.2.3 Voltage regulation at minimum temperature-altitude conditions. The above test (4.7.2.2) shall be repeated at minimum temperature-altitude conditions except that the initial, non-operating soak time at the initial, minimum temperature, sea level condition shall be increased to 2 hours minimum.

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4.7.2.4 Operating position. The converter shall be operated at 12 volts input full load, in the normal base down position until output voltage stabilizes. Output voltage, and critical component temperatures shall be measured as the converter is rotated by increments of 90 degrees from the normal base down position,360 degrees about each major axis except the vertical axis. Each position shall be held until output voltage, and critical component temperatures stabilize.

4.7.2.5 Worst condition. The converter shall be operated at the worst indicated conditions determined from the tests specified in 4.7.2.1 through 4.7.2.4. The conditions from each of these tests resulting in highest output voltage shall be simultaneously applied. Prior to the test, the converter shall be soaked at the applicable ambient temperature for a minimum of 2 hours prior to energizing the converter. The converter shall be operated until voltage, and critical component temperatures stabilize. The above test shall be repeated for minimum voltage, and maximum critical component temperature.

4.7.3 Efficiency. The converter efficiency shall be computed from data taken during tests specified in 4.7.2.1. The efficiency shall be calculated under these conditions and shall be no less than specified in 3.6.3. When computing efficiency, mean (average) values of input and output voltage and current shall be used.

4.7.4 Recovery time. The output voltage transient shall be measured when the full-rated load is applied and removed at input voltages of 9, 20 and 32 volts. The unit shall not develop sustained oscillations and the output voltage shall remain within the limits specified in 3.6.6. This test shall be conducted at minimum, maximum and room ambient temperatures specified, at approximately sea level altitude.

4.7.5 Starting.

4.7.5.1 Starting Voltage. The output voltage shall be measured when the converter is energized at 9, 20, and 32 volts at zero, one half and full load conditons. Output voltage shall meet the requirements of 3.6.7. This test shall be conducted at minimum, maximum and room ambient temperature specified, at approximately sea level altitude.

4.7.5.2 Starting Current. The input starting current shall be measured while starting the unit with half and full load at 9,

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20,and 32 volts input. The starting current measurements shall be conducted at minimum, maximum and ambient temperatures specified at approximately sea level altitude.

4.7.6 Overload. The converter shall be tested to demonstrate compliance with the overload capacity requirements specified in 3.6.2. The overload test shall be conducted at maximum, minimum and ambient temperatures specified at approximately sea level pressure.

4.7.7 Electromagnetic Compatibility. The converter shall meet the requirements of MIL-STD-461A, Notice 1 for Class IIIA equipment when tested in acccrdance with MIL-STD-462, Notice 1. The converter shall be tested at zero, half and full load conditions at 12 and 32 volts input.

4.7.8 Output Ripple Voltage. Output ripple voltage shall be measured during the tests specified in paragraph 4.7.2.1, 4.7.2.2 and 4.7.2.3 to determine compliance with 3.6.4.

4.7.9 Influence on electric system. The converter shall be tested to verify compliance with 3.5.2.2. Tests shall be conducted at no load, half load and full load and 9, 20, and 32 volts input.

4.7.10 Temperature-altitude cycling. The converter shall be operated at full rated load, at the temperature-altitude conditions indicated.

		CLASS A CONVERTER		CLASS B CONVERTER	
Step	Test Time Hrs	Altitude Feet	Temperature °C	Altitude Feet	Temperature °C
1	5	sea level	-55	sea level	-55
2	5	sea level	71	17,500	85
3	5	30,000	0	40,000	40
4	5	50,000	-10	65,000	20

The test shall consist of a total of 4 continuous cycles, each cycle consisting of steps 1 through 4. The input voltage shall be 12 VDC, except that during step 2, the input voltage shall be adjusted to 9 VDC for one minute. The output voltage shall be monitored during the entire test for compliance with the limits specified in 3.6.1.1. After each cycle, the voltage regulation shall be measured as specified in 4.7.2.1. The air flow and temperature within the test chamber shall be regulated and maintained as specified in 4.6.5.

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4.7.11 Input power susceptibility.

4.7.11.1 Transient spike susceptibility. The converter shall be operated at no load and full load, at an input voltage of 28 VDC. An input transient spike of +600 volts, containing one joule of energy shall then be applied to the input of the converter. With the input voltage removed, an input transient spike of -600 volts, one joule of energy shall be applied to the input of the converter. Each transient shall be applied 4 times for each load condition. The 600 volt, one joule spike can be conveniently obtained by the discharge through the input of the converter of a 5.55 mfd capacitor fully charged to 600 volts (Energy (joules) = $\frac{1}{2}CE^2$, where C is capacitance in farads and E is the charge on the capacitor in volts). The transient source (such as the capacitor) shall be connected to the converter by 14 AWG (or larger gage) wire, not more than 12 inches in length. The converter shall be observed for evidence of flashover or insulation breakdown and the output shall be monitored to verify compliance with 3.6.1.3. Subsequent to the test, the output voltage and frequency regulation shall be measured as specified in 4.7.2.1.

4.7.11.2 Transient surge susceptibility. With the converter operating at no load and full load with an input of 28 VDC, a positive step voltage of 80 volts with a duration of 50 milliseconds shall be applied to the converter input. The transient surge shall be obtained from a zero inpedance source, such as a capacitor bank. The transient surge shall be applied four consecutive times for each load condition. The magnitude and duration shall be monitored on an oscilloscope and shall be measured, prior to application to the converter under test, by applying and adjusting the transient across a dummy load which approximates the input of the converter under test. The converter shall be observed for evidence of flashover or insulation breakdown and the output shall be monitored to verify compliance with 3.6.1.3. Subsequent to the test, the output voltage regulation shall be measured as specified in 4.7.2.1.

4.7.11.3 Ripple voltage susceptibility. With the converter operating at no load, 1/4, 1/2, 3/4 and full rated load at input voltages of 12, 20, and 32 volts, an audio frequency signal shall be superimposed on the converter input. The amplitude and frequency of the signal shall conform to the curve shown in Figure 7 of MIL-STD-704. The input and output of the converter shall be monitored during the test in order to measure output voltage, and ripple to verify compliance to 3.6.1.1 and 3.6.4.

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4.7.11.4 Source Impedance. The test of paragraph 4.7.2.1 shall be conducted with resistance added in series with the input of the converter to give the specified source impedance (see 3.2).

4.7.12 Acceleration. The converter shall be tested in accordance with MIL-E-81910(AS) Procedure I and II. The converter shall be operated at full load, 20 VDC input voltage during procedure II. The output voltage shall be monitored during the test to determine compliance with 3.7.8. At the conclusion of each test procedure, the converter shall be inspected for evidence of damage and voltage regulation shall be measured as specified in 4.7.2.1.

4.7.13. Humidity. The converter shall be tested in accordance with MIL-E-81910(AS). The converter shall not be electrically energized during the test. Following the test, the voltage regulation shall be measured as specified in 4.7.2.1. The converter shall be inspected for evidence of damage.

4.7.14 Shock. The converter shall be tested in accordance with MIL-E-81910(AS). The converter shall be operated at full load, 20 volts input during the basic design portion of the test. Following this portion of the test, voltage regulation shall be measured as specified in 4.7.2.1.

4.7.15 Explosive atmosphere. (as required) - When specified (see 3.2), the converter shall be tested in accordance with MIL-STD-810, Method 511, Procedure I.

4.7.16 Vibration. The converter shall be vibrated in accordance with MIL-E-81910(AS). The converter shall comply with 3.7.6 while operating at full load, 20 volts input during the test. Following the test, voltage regulation shall be measured as specified in 4.7.2.1.

4.7.17 Fungus. The converter shall be tested in accordance with MIL-E-81910(AS). Operation during the test shall not be required. At the conclusion of the test, the converter shall be inspected for fungus growth and the voltage regulation shall be measured as specified in 4.7.2.1.

4.7.18 Salt fog. The converter shall be tested in accordance with MIL-E-81910(AS). At the conclusion of the test, the voltage regulation shall be measured as specified in 4.7.2.1.

4.7.19 Dust. The converter shall be tested in accordance with MIL-E-81910(AS). The converter shall be operated at full

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load with an input voltage of 20 VDC, during operational portions of the test and shall comply with 3.7.3. At the conclusion of the test, the voltage regulation shall be measured as specified in 4.7.2.1.

4.8 Reliability demonstration. The reliability demonstration test shall consist of the tests specified in 4.8.1 and 4.8.2. At the option of the manufacturer, the reliability demonstration test may be preceded by the burn-test of 4.8.3. Accept/reject criteria shall be as specified in 4.8.4.

4.8.1 Endurance (cycling test). The converter shall be operated for a total of 400 hours under full rated load at room ambient temperature and altitude.

a. During the first 200 hours the converter shall be subjected to continuous cycles of operation consisting of one minute at 9 volts input at full load and four minutes at 28 volts input and full load. Following this test the voltage regulation shall be measured in accordance with 4.7.2.1. Voltage during the test shall be within the limits specified in 3.6.1.1.

b. During the last 200 hours, the input voltage shall be alternately switched on and off for 5 second periods with full load, 20 volts input. Following this test, the voltage regulation shall be measured in accordance with 4.7.2.1 voltage shall be within the limits specified by 3.6.1.1. During the entire test period, output voltage measurements shall be taken at approximately 24 hour intervals and shall be within the limits specified by 3.6.1.1.

4.8.2 Life. The converter shall be operated for a total of 2000 hours, at full rated load and 20 VDC input voltage. For at least 48 hours of the total test time, the converter shall be continuously operated at sea level and maximum ambient temperature for the class of converter being tested. Operation for the remaining test time shall be at room temperature. Output voltage measurements shall be taken at the conclusion of the 48 hour high temperature operation while the converter is at the high temperature. During the room temperature test period, output voltage shall be measured at approximately 250 hour intervals. At the conclusion of the entire test, voltage regulation and ripple voltage shall be measured as specified in 4.7.2.1 and 4.7.8.

4.8.3 Burn-in. The converter shall be continuously operated for a minimum of 48 hours at full rated load 20 VDC input voltage

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at room ambient conditions. Output voltage shall be monitored during the test and measured at the conclusion of the test to detect any deviation of output voltage in excess of that specified in 3.6.1.1.

4.8.4 Failure definition. Any voltage measurement during or after the tests of 4.8.1, 4.8.2 and 4.8.3 not within the limits specified in 3.6.1.1 shall constitute a failure. Failure to comply with the requirements specifiied in 3.6.4 after the test of 4.8.2 shall also consitute a failure. Any converter that fails to comply with specified requirements during the test of 4.8.3 may be repaired and returned to test for an additional 48 hours. The retest shall be completed without failure. Any converter that fails during the retest shall not be delivered on any contract or order.

Compliance with the reliability requirement specified in 3.8 is demonstrated by two converters completing the test of 4.8.1 and 4.8.2 without failure.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The converter shall be unit packaged in accordance with Military Standard MIL-STD-794, level A or C as specified in contract or order.

5.1.1. Level A. For level A packaging, preservation shall be in accordance with Specification MIL-P-116, Method II, without preservation compounds, using metal reusable containers unless otherwise specified.

5.1.2 Level C. For level C the converter shall be preserved and packaged individually in accordance with manufacturer's commercial practice.

5.2 Packing. Packing shall be in accordance with MIL-STD-794 level A, B or C as specified in contract or order.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-794 and Military Standard MIL-STD-129.

5.3.1 Reinspection dates. Reinspection date marking shall be as specified by the procuring activity.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type

c. Military part number

d. Required level of packaging and packing

e. Type of container

f. Reinspection date marking and any special marking

g. Installation instructions, handbooks, or manuals when required

6.2 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for Qualification in order that they may be eligible to be awarded contracts or orders for the products coverted by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC, and information pertaining to qualification of products may be obtained from this activity.

6.2.1 Reproducible copies. Upon completion of qualification testing and prior to qualification approval, the manufacturer shall provide two reproducible copies of outline and detail assembly drawings, schematics and parts list. Any changes in the reproducible copies from those submitted with the qualification samples shall be indicated in detail.

6.3 Installation. The aircraft manufacturer shall allow clearance of at least one inch at each end of the converter in addition the maximum dimensions shown on the detail specification for installation and ventilation.

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Navy - AS

Project No. 6130-N-157

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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